# University Orientation System using Classical Reasoning, Default Reasoning and Fuzzy Logic

*Final Project (TP-RCR) — Knowledge Representation and Reasoning*

4th year Ingéniorat IA — University of Guelma, 2026/2027

## 1. Introduction and objective

We build a **University Orientation System** that decides, for a student profile (grades, skills, interests), which of three majors — **Artificial Intelligence (AI)**, **Software Engineering (SE)** and **Cybersecurity (CYB)** — are *accessible* (eligibility) and *recommended* (orientation). The same knowledge base is reasoned upon with three methods of the syllabus and the results are compared:

1. **Classical logic** (Chapitre I-A): monotonic deduction by forward chaining over Horn clauses;
2. **Default logic** (Chapitre I-B.2): non-monotonic reasoning — Reiter defaults `α : β / γ`, extension computation, defeasible presumptions and effective belief retraction;
3. **Fuzzy logic** (Chapitre III): Mamdani fuzzy inference with Low/Medium/High membership functions producing exact suitability percentages.

Everything is implemented in **pure Python 3 (standard library only)**: `run_all.py` executes the five steps and writes all logs/JSON to `output/`; 22 unit tests (`tests/test_system.py`) pin every claim made below, so each conclusion is justified by executable code and data.

## 2. Step 1 — Knowledge modeling

### 2.1 Majors and orientation criteria

| Major | Key subjects | Accessibility (eligibility) | Recommendation adds |
|---|---|---|---|
| **AI** | Mathematics, Algorithmics | algo ≥ 14 **and** math ≥ 14 | declared interest ≥ 7 **or** problem-solving ≥ 7 |
| **SE** | Programming, Algorithmics | prog ≥ 14 **and** average ≥ 12 | declared interest ≥ 7 **or** coding practice ≥ 7 |
| **CYB** | Networks, Programming | networks ≥ 14 **and** average ≥ 12 | declared interest ≥ 7 **or** problem-solving ≥ 7 |

Each student is described by **grades** (math, algorithmics, programming, networks, on 0–20), **skills** (problem_solving, coding_practice, on 0–10) and **interests** (ai, se, cyber, on 0–10; the whole questionnaire may be *missing* = `null`). Thresholds (`data/criteria.json`): high grade ≥ 14, good average ≥ 12, low average < 10, low math ≤ 9, strong skill ≥ 7, positive interest ≥ 7, negative interest (dislike) ≤ 3, tie ≥ 7.

### 2.2 Shared fact vocabulary (grounding predicates into propositions)

`src/common.derive_facts()` grounds first-order predicates on one student into propositional atoms (the fact base **W** shared by the classical and default engines):

| Predicate (form) | Ground atom | Derived iff |
|---|---|---|
| HighAlgo(s), HighMath(s), HighProg(s), HighNet(s) | `high_algorithmics`, … | grade ≥ 14 |
| GoodAvg(s) / LowAvg(s) | `good_average` / `low_average` | avg ≥ 12 / avg < 10 |
| WeakMath(s) | `low_math` | math ≤ 9 |
| StrongPS(s) / StrongCoding(s) | `strong_problem_solving` / `strong_coding` | skill ≥ 7 |
| Interested(s,X) / Dislikes(s,X) | `interests_x` / `dislikes_x` | interest ≥ 7 / ≤ 3 |
| UnknownInterest(s,X) | `interest_unknown_x` | questionnaire missing for X |
| Tied(s) | `tied_se_cyber` | interest_se = interest_cyber ≥ 7 |

A **missing grade or interest yields no positive/negative atom at all**: classical logic then treats it as false (closed world), whereas default logic can exploit the explicit `interest_unknown_x` markers. This dual treatment is the backbone of the comparison.

### 2.3 The ten student profiles (`data/students.json`)

| ID | Student | math | algo | prog | net | ps | code | interests (ai/se/cyb) | Designed phenomenon |
|---|---|---|---|---|---|---|---|---|---|
| S01 | Amina Belkacem | 17 | 18 | 15 | 12 | 9 | 6 | 9 / 5 / 3 | clear AI talent, dislikes cyber |
| S02 | Yacine Haddad | 12 | 13 | 17 | 10 | 7 | 9 | 5 / 9 / 4 | clear SE profile |
| S03 | Meriem Saadi | 11 | 10 | 12 | 16 | 7 | 5 | 2 / 3 / 9 | clear CYB profile |
| S04 | Sofiane Meziane | 9 | 8 | 10 | 9 | 4 | 4 | 6 / 5 / 5 | weak record → remediation |
| S05 | Lina Zerouki | 15 | 16 | 14 | 11 | 6 | 5 | **missing** | missing data: abstain vs presume vs impute |
| S06 | Karim Boudjema | 16 | 15 | 14 | 10 | 6 | 7 | 1 / 8 / 2 | **exception**: AI grades but hates AI |
| S07 | Ines Larbi | 15 | 17 | 15 | 9 | 8 | 6 | **missing**, then 2 / 8 / 3 | **belief revision** (late questionnaire) |
| S08 | Omar Cherif | 11 | 12 | 13 | 17 | 7 | 6 | 3 / 4 / 8 | solid CYB, borderline for strict SE |
| S09 | Nadia Slimani | 13 | 14 | 14 | 13 | 6 | 6 | 6 / 6 / 6 | **borderline**: threshold artefacts |
| S10 | Riad Amrani | 14 | 14 | 15 | 15 | 8 | 8 | 5 / 7 / 7 | **ambiguity/tie** SE = CYB |

## 3. Step 2 — Classical logic reasoning (Chapitre I-A)

### 3.1 Formalization and rule base

Knowledge is written as Horn clauses, e.g. C1 in predicate logic `∀s: EligibleAI(s) ← HighAlgo(s) ∧ HighMath(s)`, grounded per student into `eligible_ai ← high_algorithmics ∧ high_math`. `data/rules_classical.json` contains:

| # | Rule | # | Rule |
|---|---|---|---|
| C1 | `eligible_ai ← high_algorithmics ∧ high_math` | C6 | `recommended_se ← eligible_se ∧ interests_se` |
| C2 | `eligible_se ← high_programming ∧ good_average` | C7 | `recommended_se ← eligible_se ∧ strong_coding` |
| C3 | `eligible_cyber ← high_networks ∧ good_average` | C8 | `recommended_cyber ← eligible_cyber ∧ interests_cyber` |
| C4 | `recommended_ai ← eligible_ai ∧ interests_ai` | C9 | `recommended_cyber ← eligible_cyber ∧ strong_problem_solving` |
| C5 | `recommended_ai ← eligible_ai ∧ strong_problem_solving` | C10 | `needs_remediation ← low_average` |

### 3.2 Deduction engine

`src/classical_logic.py` implements **forward chaining**: starting from W, any rule whose body is satisfied is fired by *Modus Ponens*, its head accumulates, and passes repeat until a **fixpoint**. Properties: sound and complete for Horn KBs, terminating (each head is added once), and **monotonic** — facts are never removed. Missing data follows the **Closed World Assumption** (underivable = false). Every firing is traced and `explain()` rebuilds a **justification tree** (and a *near-miss* analysis for non-derived atoms: which body atom was missing).

### 3.3 Results on the 10 profiles (extract of `output/run_log.txt`)

| Student | Eligible | Recommended | Justification highlight |
|---|---|---|---|
| S01 | AI, SE | **AI** | C1+C4: algo 18 ∧ math 17 ∧ interest 9 |
| S02 | SE | **SE** | C2+C6: prog 17 ∧ avg 13 ∧ interest 9 |
| S03 | CYB | **CYB** | C3+C8: net 16 ∧ avg 12.25 ∧ interest 9 |
| S04 | – | – (+needs_remediation) | C10: avg 9 < 10 |
| S05 | AI, SE | **– (abstains)** | C4 fails: `interests_ai` MISSING (questionnaire null) |
| S06 | AI, SE | **SE** | C4 blocked (interest ai = 1); C6 fires (interest se = 8) |
| S07 (phase 1) | AI, SE | **AI** | C5 via `strong_problem_solving` = 8 |
| S08 | CYB | **CYB** | C3+C8 |
| S09 | SE | **– (abstains)** | math 13 < 14 kills C1; interest/skills 6 < 7 kill C4–C9 |
| S10 | AI, SE, CYB | **AI, SE, CYB** | C5+C6+C8 — three recommendations, no ranking |

**Observed limitations.** (i) *Abstention on missing data*: S05 is eligible everywhere her grades allow but never recommended — the near-miss trace shows exactly `interests_ai MISSING`. (ii) *Threshold artefacts*: S09 loses AI eligibility for **one point** of mathematics (13 vs 14). (iii) *Monotonicity*: when S07's follow-up questionnaire arrives (she declares interest ai = 2), classical logic still recommends AI (via C5) — nothing can be retracted. (iv) *No ranking*: S10 gets three recommendations with no preference mechanism.

## 4. Step 3 — Default logic reasoning (Chapitre I-B.2)

### 4.1 Default theory

A Reiter default `α : β₁,…,βₙ / γ` applies when its prerequisite α is believed and each justification $\beta_i$ is *consistent* with the final belief set ("**γ normally holds, unless data proves otherwise**"). `data/rules_default.json` defines **8 defaults** (≥ 4 required) plus 7 strict rules (S1a…S4, encoded as justification-free defaults `α : / γ` so they hold in every extension):

| # | Default α : β / γ | Reading |
|---|---|---|
| D1 | `high_algorithmics : ¬dislikes_ai, ¬low_math / suited_ai` | strong in algo ⇒ normally suited for AI, **unless** declared dislike or critically weak math |
| D2 | `high_programming : ¬dislikes_se / suited_se` | strong programmer ⇒ normally suited for SE, unless dislike |
| D3 | `high_networks : ¬dislikes_cyber / suited_cyber` | strong in networks ⇒ normally suited for CYB, unless dislike |
| D4a-c | `suited_X ∧ interest_unknown_X : ¬dislikes_X / recommended_X, presumed_interest_X` | **missing questionnaire ⇒ presume a compatible interest** and recommend, unless dislike |
| D5a | `suited_se ∧ suited_cyber ∧ tied_se_cyber : ¬oriented_se / oriented_cyber` | tie ⇒ orient CYB unless oriented SE |
| D5b | `suited_se ∧ suited_cyber ∧ tied_se_cyber : ¬oriented_cyber / oriented_se` | tie ⇒ orient SE unless oriented CYB |

Strict rules turn aptitude into recommendation on *declared* evidence, e.g. S1a `suited_ai ∧ interests_ai : / recommended_ai`, S1b `suited_ai ∧ strong_problem_solving : / recommended_ai` (idem S2a/S2b for SE, S3a/S3b for CYB), S4 `low_average : / needs_remediation`.

### 4.2 Extension computation

`src/default_logic.py` computes **all extensions** by generate-and-test over *processes* (§2.3 "Calcul des extensions" of the syllabus): a DFS appends only defaults applicable to the accumulated belief set (early pruning), maximal processes are validated against their **final** belief set (prerequisites still believed, justifications still consistent), and duplicates are merged. Example sizes: S09 → 5 processes explored → 1 extension; **S10 → 65 335 processes → 2 extensions**. Each conclusion keeps its *support* (which defaults produced it), which is what makes retractions and re-justifications observable.

### 4.3 Results, exception, and effective belief revision

| Student | #Ext | Suited (presumed aptitude) | Recommended | Notes |
|---|---|---|---|---|
| S01 | 1 | AI, SE | AI | D3 blocked twice (net 12 < 14; `dislikes_cyber`) |
| S02 | 1 | SE | SE | aligned with classical |
| S03 | 1 | CYB | CYB | D1/D2 defeated by `dislikes_ai`, `dislikes_se` |
| S04 | 1 | – | – (+ remediation) | S4 strict |
| S05 | 1 | AI, SE | **AI*, SE*** | * presumed by D4a/D4b — classical abstained |
| S06 | 1 | SE | SE | **exception**: D1 defeated by `dislikes_ai` despite algo 15, math 16 |
| S07 ph.1 | 1 | AI, SE | AI, SE* | AI presumed (D4a) *and* strict (S1b); SE presumed (D4b) |
| S08 | 1 | CYB | CYB | aligned with classical |
| S09 | 1 | **AI, SE** | – | presumes AI aptitude although crisp C1 rejected it (math 13) |
| S10 | **2** | AI, SE, CYB | AI, SE, CYB | E1 concludes `oriented_cyber` (D5a), E2 `oriented_se` (D5b) — Nixon-diamond ambiguity |

**Exception (required).** For S06 the fact `dislikes_ai ∈ W` (interest ai = 1 ≤ 3) makes D1's justification `¬dislikes_ai` inconsistent: `suited_ai` is **never concluded**, although the student has the best AI grades of the cohort. The same mechanism defeats D1/D2/D3 for S03.

**Belief revision (required) — S07, two phases.** Phase 1: questionnaire missing ⇒ $W_1 \supseteq$ {high_algorithmics, high_math, high_programming, strong_problem_solving, interest_unknown_ai/se/cyber}; the single extension concludes {suited_ai, suited_se, recommended_ai, presumed_interest_ai, recommended_se, presumed_interest_se}. Phase 2: the follow-up arrives (ai = 2, se = 8, cyber = 3) ⇒ $W_2 = W_1 \cup$ {dislikes_ai, interests_se, dislikes_cyber} {interest_unknown_*}. Re-running the engine:

- **RETRACTED**: `suited_ai`, `recommended_ai`, `presumed_interest_ai`, `presumed_interest_se` (D1 defeated by `dislikes_ai`; D4a/D4b lose their prerequisites);
- **KEPT but re-justified**: `recommended_se` — support switches from presumption **D4b** to strict evidence **S2a** (`interests_se` = 8 now declared);
- Formally **$Cn(W_1) \not\subseteq Cn(W_2)$**: non-monotonicity is *effective*, whereas classical logic keeps recommending AI to S07 forever (monotonicity check `derived(W₁) ⊆ derived(W₂)` = True in `output/revision_demo.json`).

**Comparison with the classical conclusions.** On complete, unambiguous profiles (S01–S04, S08) both logics agree. They diverge exactly where knowledge is incomplete or exceptional: missing data (S05, S07-ph1: classical *abstains*, defaults *presume* — traceably, with `presumed_interest_*`), exceptions (S06: defaults *refuse* an aptitude that grades alone would suggest), borderline values (S09: D1 presumes `suited_ai` where the crisp conjunction C1 rejects), and ambiguity (S10: default logic *represents* the hesitation as two extensions instead of hiding it).

## 5. Step 4 — Fuzzy logic reasoning (Chapitre III)

### 5.1 Fuzzy variables and membership functions

`data/fuzzy_config.json` defines four linguistic variables (figure: `output/figures/membership_functions.svg`). The three inputs form **Ruspini partitions** (μ_Low+μ_Medium+μ_High = 1 at every point — verified by unit test):

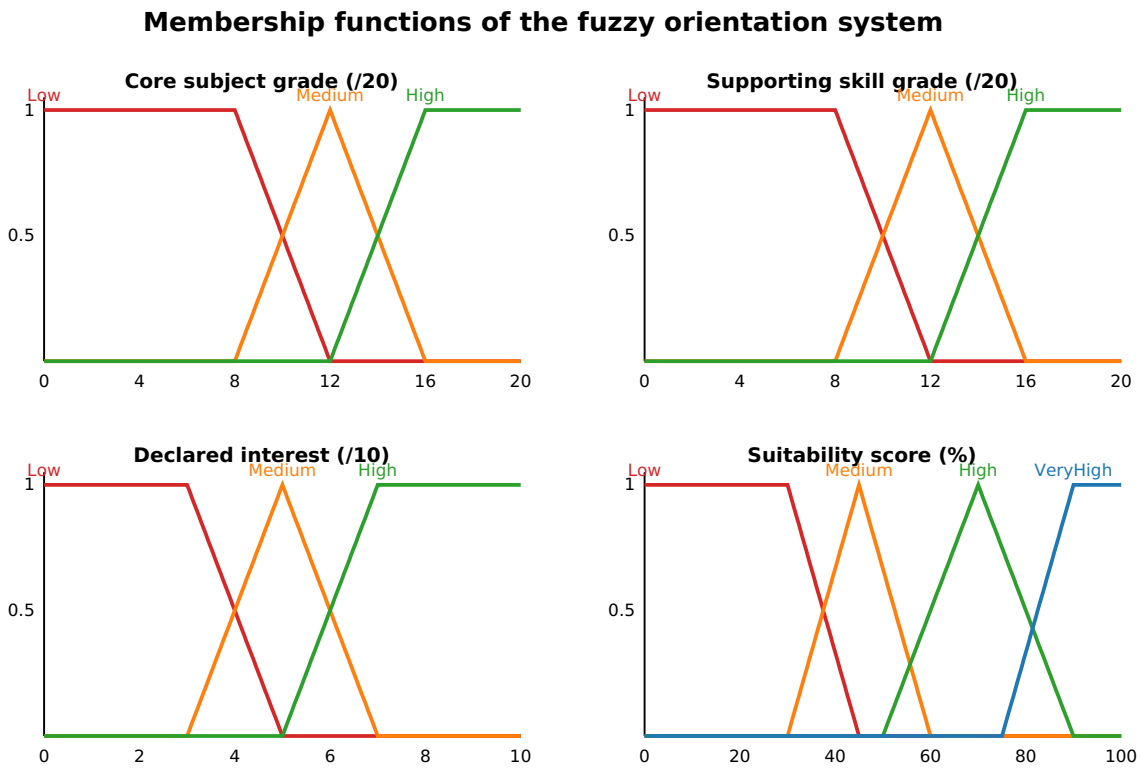



*Figure 1 — Membership functions of the four linguistic variables (generated by src/fuzzy_logic.render_membership_svg)*

| Variable | Domain | Low | Medium | High | VeryHigh |
|---|---|---|---|---|---|
| **Core grade** (key subject) | [0,20] | trap(0,0,8,12) | tri(8,12,16) | trap(12,16,20,20) | – |
| **Skill grade** (support subject) | [0,20] | trap(0,0,8,12) | tri(8,12,16) | trap(12,16,20,20) | – |
| **Interest** | [0,10] | trap(0,0,3,5) | tri(3,5,7) | trap(5,7,10,10) | – |
| **Suitability** (output) | [0,100] % | trap(0,0,30,45) | tri(30,45,60) | tri(50,70,90) | trap(75,90,100,100) |

Per major the same rule base is instantiated by an input mapping: **AI** (core = math, skill = algorithmics, interest = ai), **SE** (core = programming, skill = algorithmics, interest = se), **CYB** (core = networks, skill = programming, interest = cyber).

### 5.2 Fuzzy rule base (13 rules ≥ 6 required)

FR1 Core H ∧ Skill H → High · FR2 Core H ∧ Int H → High · FR3 Core M ∧ Skill M → Medium · FR4 Skill H ∧ Int M → High *(the TP example)* · FR5 Core L → Low · FR6 Skill L ∧ Int L → Low · FR7 Core M ∧ Int H → Medium · FR8 Core H ∧ Skill L → Medium · FR9 Core L ∧ Skill H → Low · FR10 Core H ∧ Int L → Medium · FR11 Int L ∧ Skill H → Low · FR12 Core M ∧ Skill L → Low · FR13 Core H ∧ Skill H ∧ Int H → VeryHigh.

FR9/FR11 encode “talent cannot compensate a failing core subject / absent motivation”; FR10 lets grades partially compensate low motivation; FR13 distinguishes full excellence (it removes the saturation artefact of a single High shoulder). The rule base is **complete**: every input combination fires at least one rule (checked by enumeration over the partitions).

### 5.3 Inference and exact suitability percentages

`src/fuzzy_logic.py` implements the Mamdani pipeline: fuzzification → rule evaluation with **AND = min** → firing strengths $\alpha_r$ → aggregation **$\mu_{agg}(y) = \max_r \min(\alpha_r, \mu_{B_r}(y))$** → **centroid defuzzification** (numerical integration, 401 samples). A missing interest is imputed to the neutral value **5.0** (full Medium membership) and flagged in the trace. Decision rule: the top major is recommended iff its score ≥ 60 %.

| Student | AI % | SE % | CYB % | Fuzzy decision | Classical said | Default said |
|---|---|---|---|---|---|---|
| S01 | **79.5** | 70.0 | 23.2 | AI | AI | AI |
| S02 | 54.5 | **73.2** | 41.4 | SE | SE | SE |
| S03 | 28.1 | 28.1 | **70.0** | CYB | CYB | CYB |
| S04 | 23.2 | 20.6 | 23.2 | none (< 60 %) | remediation | remediation |
| S05 | **70.0** | **70.0** | 48.9 | AI = SE (tie) | abstains | AI*, SE* |
| S06 | 40.4 | **67.1** | 28.1 | SE | SE | SE |
| S07 ph1 | **70.0** | **70.0** | 39.6 | AI = SE (tie) | AI | AI, SE* |
| S08 | 34.4 | 45.0 | **73.2** | CYB | CYB | CYB |
| S09 | 63.4 | **67.1** | 63.4 | SE | abstains | – (suited AI, SE) |
| S10 | 59.5 | 70.8 | **73.2** | CYB | 3 recs, no rank | 2 extensions |

**Worked example (S09 → AI, from `output/fuzzy_results.json`).** Inputs: core = math 13 (μ_M = 0.75, μ_H = 0.25), skill = algo 14 (μ_M = 0.5, μ_H = 0.5), interest = 6 (μ_M = 0.5, μ_H = 0.5). Fired rules: FR3 α = 0.5 → Medium; FR4 α = 0.5 → High; FR7 α = 0.5 → Medium; FR1 α = 0.25 → High; FR2 α = 0.25 → High; FR13 α = 0.25 → VeryHigh. Aggregation + centroid ⇒ **63.4 %**: where the crisp threshold (math 13 < 14) rejected AI outright, fuzzy logic answers “moderately suited” — gradation instead of a cliff. Revision (S07): when interests (2/8/3) replace the neutral imputation, AI falls 70.0 % → **37.4 %** while SE rises 70.0 % → **73.2 %**: revision by smooth re-evaluation, without symbolic retraction.

## 6. Step 5 — Comparative analysis

| Dimension | Classical logic | Default logic | Fuzzy logic |
|---|---|---|---|
| Knowledge form | Horn clauses (crisp True/False) | Defaults α:β/γ + strict rules | Fuzzy rules over linguistic terms |
| Truth values | 2-valued | 2-valued but *provisional* | degrees μ ∈ [0,1] |
| Mechanism | Forward chaining, Modus Ponens, fixpoint | Extension computation (processes), justification consistency | Mamdani: min/max + centroid |
| **Missing data** | CWA ⇒ **abstention** (S05: eligible, never recommended) | **Presumption** D4a-c ⇒ actionable answer, traceable (`presumed_interest_*`) and revocable | **Neutral imputation** (interest = 5 ⇒ Medium) ⇒ always a ranked score |
| **Exceptions** | Cannot express "normally"; strict rules only | Justifications **defeat** defaults (S06: `dislikes_ai` blocks D1) | No exceptions; counter-evidence just lowers the score smoothly (S06 AI 40.4 % vs ≈70 % potential) |
| **Revision** | **Monotonic** — never retracts (S07 keeps AI recommendation despite declared dislike) | **Non-monotonic** — effective retraction (S07: `suited_ai`, `recommended_ai` withdrawn; `recommended_se` re-justified D4b→S2a) | Re-evaluation: scores move continuously (AI 70→37.4 %) |
| **Ambiguity/ties** | Recommends everything that matches, no ranking (S10) | **Multiple extensions** make the hesitation explicit (S10: 2 extensions; choice = human decision) | Close scores give a soft quantitative tie-break (73.2 vs 70.8 vs 59.5) |
| **Imprecision** | Crisp thresholds ⇒ artefacts at boundaries (S09: one point of math flips eligibility) | Crisp, but presumptions soften rejections (S09: `suited_ai` presumed) | Native: gradual membership (μ_High(13) = 0.25) |
| Explanation | Proof trees + near-miss analysis | Supports per conclusion, applied defaults per extension | Fired rules with α, membership degrees |
| Cost / complexity | O(rules × passes) — trivial | Exponential in theory (S10: 65 335 processes); fine here | O(rules × samples) — trivial |
| Main pros | Simple, sound & complete, fully explainable | Commonsense "unless" reasoning; revisable; represents alternatives | Robust to imprecision; always ranks; tolerates missing values |
| Main cons | Brittle thresholds; abstains; cannot retract or rank | Extension multiplicity costly; conclusions hard to rank; choice among defaults is a design decision | Scores not symbolic — no explicit retraction; MF/rule design is subjective |

**Synthesis.** The three approaches are complementary rather than competing: classical logic gives the *administrative floor* (auditable eligibility), default logic adds *commonsense flexibility* (presume, defeat, revise — the only one that retracts S07's recommendation when the exception arrives), and fuzzy logic provides the *graded ranking* the other two lack (it separates S10's three-way tie and rescues S09 from a one-point rejection). A realistic orientation service would pipeline them: crisp eligibility → defeasible recommendation with revision → fuzzy scoring for the final ranking, with the symbolic traces as the explanation layer.

## 7. Conclusion and validation map

All five TP steps are implemented and exercised on the 10 profiles: (1) a shared knowledge model with three majors, ten profiles and an explicit fact vocabulary; (2) a classical forward-chaining engine whose every conclusion (and refusal) is justified by a proof/near-miss trace; (3) a default theory with 8 defaults, full extension computation, one effective exception (S06) and one effective retraction (S07), plus a two-extension ambiguity case (S10); (4) a from-scratch Mamdani system with 4 variables, 13 rules and centroid-defuzzified percentages; (5) the comparative analysis above, grounded in the observed divergences. For the validation session: `output/run_log.txt` and the four JSON result files contain every number quoted here, and `python3 -m unittest discover -s tests` (22 tests, all green) re-proves each claim directly from the code and data.